Core Foundations (Start Here)

These modules lay the conceptual and architectural groundwork.

Module 1: Foundations of Agentic AI & Generative Intelligence

- Evolution: From LLMs to Agents
- What makes an agent *agentic*?
- Components of an intelligent agent: Planning, Memory, Tool Use, Goals
- Agent Types: Reflex, Goal-Based, Utility-Based, Learning Agents
- Generative AI Landscape: Text, Vision, Speech, Code

Assignment: Compare Agent architectures across LangChain, CrewAI, AutoGen

Module 2: Prompt Engineering & Autonomous Reasoning

- Structured prompting strategies (ReAct, CoT, ToT)
- Role of memory and dynamic prompt generation
- Self-reflection and self-evaluation strategies
- Toolformer-style reasoning

Mini Project: Build a smart travel planner that reasons across multiple goals and reflects on its suggestions

Core Development Skills (Build the Brain)

Build hands-on capabilities to construct and scale agentic systems.

Module 3: RAG & Multi-modal Systems

- Vector Databases: FAISS, Pinecone, Weaviate
- Chunking & Embedding Optimization
- RAG with PDFs, CSVs, YouTube, Codebases
- Multi-modal RAG (text+image+video): Architecture and applications

Project: Build a Document Q&A + YouTube + Chart reader RAG system

Module 4: Agentic Frameworks Deep Dive

- LangChain, AutoGen, CrewAI, LangGraph, MetaGPT
- Planning and Scheduling
- Memory: Episodic, Long-Term, Retrieval-Based
- Tool Use and Tool Chaining
- Feedback loops, retries, exception handling

Assignment: Convert a prompt-based app into an autonomous agent with task decomposition

Module 5: Multi-Agent Collaboration & Communication

- Architecting multi-agent workflows

- Roles: Manager, Worker, Evaluator, Researcher
- Agent communication protocols
- Parallel execution and delegation strategies

Project: Create a multi-agent research team (Researcher + Writer + Editor + Reviewer)

Real-World Execution (Bring to Life)

Deploy, integrate, and make your agents interact with real-world environments.

Module 6: Real-Time & Event-Driven Agents

- Event-based triggers, webhooks, socket communication
- Agents that respond to API calls, calendar updates, user commands
- Real-time planning and adaptive behavior

Project: Personal productivity assistant that responds to events (emails, calendars, chat)

Module 7: AgentOps & Scaling in Production

- Observability: Logging, analytics, telemetry
- Safety: Guardrails, filters, ethical considerations
- Deployment: Docker, FastAPI, LangServe
- Latency optimization and cost control

Assignment: Create a deployable Agent with monitoring and feedback loop

Advanced and Emerging Tracks (Next Gen AI Builders)

Specialized modules for cutting-edge agentic AI.

Module 8: Building Voice + Avatar Agents

- STT (Whisper) + TTS (Piper/ElevenLabs)
- Lip Sync with Audio2Face or Three.js Avatar
- Build Voice-Interactive Agent with Personality

Project: Develop an AI gent that can talk, express and look like actual human

Module 9: Agent Evaluation & AI Governance

- Agent Performance Metrics
- Auto-Scoring + Human-in-the-Loop
- Responsible AI, Ethics, Governance

Industry Focus & Portfolio Creation (Mastery Track)

Apply your learning in domain-specific contexts and finish with an enterprise-ready capstone.

Module 10: Enterprise Use Cases

- Finance: Autonomous reporting & reconciliation
- Healthcare: Virtual triage & medical summary generation
- Retail: Conversational agent for dynamic catalog search
- Legal: Contract analyzer with risk flagging

Module 11: Capstone Project Options:

- Autonomous AI Co-Pilot for Knowledge Work
- Dynamic Workflow Builder Agent for Task Automation
- Cross-modal Agent that interprets image + text + user input to make decisions
- 24/7 Enterprise Agent with real-time data awareness (e.g., for business dashboards)